Endocrine OSCE Cases

Question and answer review by exam year

Key: Drug names **Critical point** ★ Repeated case Answers follow ACCP Updates in Therapeutics 2026, ADA Standards of Care and ATA guidelines.

2025

Case 1: Diabetic Ketoacidosis (DKA) ★ Repeated case, also in 2022–2023, 2021, 2018–2019

Case details: Fever, tachycardia, tachypnea, sore throat; missed Insulin for 1 week; thirst and polyuria. CrCl 39 mL/min. Home: Metformin 750 mg BID, Glibenclamide 5 mg, basal-bolus Insulin. K 5.6, glucose about 350 mg/dL, bicarbonate 11, severe acidemia.

Question	Answer
Q1. Give full management of DKA in this case with doses.	• Fluids: 0.9% NaCl 15–20 mL/kg (1–1.5 L) in the first hour, then 250–500 mL/h guided by hemodynamics, urine output and corrected sodium; add dextrose 5% when glucose <200 mg/dL • Potassium (K 5.6): **do not give potassium now**; start 20–30 mEq per liter once K <5.2; goal 4–5 mEq/L • Insulin: regular Insulin 0.1 unit/kg IV bolus then 0.1 unit/kg/h (or 0.14 unit/kg/h without bolus), only after K ≥3.3; titrate to lower glucose 50–75 mg/dL/h • When glucose <200: reduce to 0.02–0.05 unit/kg/h with dextrose 5%; keep glucose 150–200 until resolved • Bicarbonate: only if pH <7.0 (100 mL of 8.4% in 400 mL water over 2 hours with potassium) • Phosphate: replace if <1 mg/dL or cardiac/respiratory dysfunction • Precipitant: sore throat and fever → cultures, treat infection • Resolution: glucose <200 plus two of bicarbonate ≥15, pH >7.3, anion gap ≤12 • Transition: SC basal-bolus (0.5–0.8 unit/kg/day) with 1–2 hour overlap before stopping the drip • Monitor: glucose hourly, BMP and venous gas every 2–4 h, ketones (β-hydroxybutyrate), K every 2 h at first, ECG, urine output, mental status • Non-pharm: NPO initially, sick-day education
Q2. What subjective and objective data help diagnose DKA?	• Subjective: thirst, polyuria, weight loss, nausea, vomiting, abdominal pain, weakness, missed Insulin, recent illness • Objective: tachycardia, hypotension, dry mucosa, Kussmaul respirations, fruity breath, altered mentation • Labs: glucose >200 mg/dL (or known diabetes), ketosis (serum or urine ketones), pH <7.30 or bicarbonate <18, high anion gap • Severity: mild pH 7.25–7.30; moderate 7.00–7.24; severe <7.00
Q3. Causes for the patient?	• Insulin omission (missed 1 week) • Infection (fever and sore throat) • Other triggers: MI, stroke, pancreatitis, steroids, SGLT2 inhibitors, new-onset diabetes
Q4. What was the reason for DKA for this patient?	• Non-adherence to Insulin, with an infection as a likely second trigger • Action: explore why (cost, fear, technique, understanding)
Q5. What class of anti-diabetic can you use upon discharge? Mention all.	• Insulin: basal-bolus (foundation) • Metformin: only with eGFR ≥30 and stable; reduce to max 1000 mg/day at eGFR 30–44 • GLP-1 receptor agonist: Semaglutide, Dulaglutide, Liraglutide • DPP-4 inhibitor: Linagliptin (no renal adjustment) • Avoid: Glibenclamide (CKD, hypoglycemia), SGLT2 inhibitors (DKA risk), Pioglitazone (fluid retention)
Q6. Semaglutide: what to screen before use?	• Personal or family history of medullary thyroid carcinoma or MEN2 • Pancreatitis history • Diabetic retinopathy (rapid A1C drop can worsen it) • Gastroparesis, gallbladder disease, severe GI disease • Pregnancy or planning; type 1 diabetes (not indicated alone) • Renal function and volume status
Q7. Dosing of Semaglutide?	• SC weekly: 0.25 mg × 4 weeks, then 0.5 mg × 4 weeks, then 1 mg; may go to 2 mg • Oral: 3 mg daily × 30 days, then 7 mg, then 14 mg (empty stomach, 30 min before food, sip of water) • Monitor: GI tolerance, hydration, weight, A1C, signs of pancreatitis
Q8. Calculate the anion gap and assess the result.	• Formula: AG = Na − (Cl + HCO3); normal 8–12 mEq/L • Example: Na 135, Cl 100, HCO3 11 → AG = 24 • Interpretation: **high anion gap metabolic acidosis** (ketoacids) • Also: correct Na for glucose (+1.6 per 100 mg/dL above 100); check delta-delta for a mixed disorder
Q9. Based on the labs, what will change for home medications upon discharge?	• Stop Glibenclamide: active metabolites accumulate in CKD (CrCl 39) → hypoglycemia • Metformin: reduce to 500 mg BID (max 1000 mg/day at eGFR 30–44); hold if acute illness • Insulin: adjust to intake and renal clearance; reinforce sick-day rules and ketone checks • Add GLP-1 RA or Linagliptin if more glucose lowering is needed • Monitor: SMBG 4×/day, A1C in 3 months, eGFR, K, B12 yearly on Metformin

Question	Answer
Also be ready for (exam add-ons)	• Goals: correct dehydration, hyperglycemia, ketoacidosis and electrolytes, treat the precipitant, then transition to SC Insulin and prevent recurrence • Non-pharm: ICU/step-down monitoring, NPO initially, strict intake/output, treat infection source, sick-day and ketone education, assess adherence barriers (cost, fear, technique) • First-line: 0.9% NaCl 15–20 mL/kg first hour; regular Insulin 0.1 unit/kg IV bolus then 0.1 unit/kg/h (only if K ≥3.3); K 20–30 mEq/L once K <5.2; Dextrose 5% when glucose <200; Bicarbonate only if pH <7.0; resolve: glucose <200 + 2 of HCO3 ≥15, pH >7.3, AG ≤12 • Pregnancy: DKA occurs at lower glucose levels (euglycemic) and is an emergency for the fetus: same protocol with fetal monitoring, left lateral position, K and phosphate care; Insulin is the only agent; SGLT2 inhibitors and GLP-1 RA stopped • Renal/dialysis: reduce Insulin needs (lower clearance), watch K and phosphate, careful fluids in oliguria; Metformin hold; Glibenclamide avoid • Liver: hypoglycemia risk (gluconeogenesis impaired); lactate interpretation • HF/CKD: fluids 250–500 mL/h guided by volume status; monitor for pulmonary edema • Counseling: never stop basal Insulin, sick-day rules, check ketones when glucose >250 or ill, SMBG 4×/day, hypoglycemia 15-15 rule, return precautions, SGLT2 inhibitors can cause euglycemic DKA • Monitoring: glucose hourly; BMP/venous gas every 2–4 h; K every 2 h at first; β-hydroxybutyrate; ECG; urine output

Case 2: DM with Hypothyroidism, CKD and Pregnancy

Case details: 35-year-old with T2DM and hypothyroidism on Metformin 1000 mg BID and Levothyroxine 100 mcg. Symptomatic for DM. A1C 8.3%. TSH high, T4 low. CKD.

Question	Answer
Q1. Is she a candidate for Insulin? Why?	• Not by A1C alone: Insulin is indicated if A1C >10%, glucose ≥300 mg/dL or catabolic features (weight loss, ketosis) • Why still consider it: symptomatic, CKD, and childbearing potential; Insulin is the preferred agent if she plans pregnancy • Now: add a second agent first, with contraception counseling
Q2. What agent will you add, with dose, and why?	• GLP-1 RA: Semaglutide 0.25 mg SC weekly × 4 weeks, then 0.5 mg, titrate to 1 mg • Why: lowers A1C and weight, kidney and CV benefit, low hypoglycemia, no renal adjustment • Alternative: SGLT2 inhibitor Empagliflozin 10 mg daily (kidney protection, eGFR ≥20) • Metformin: adjust to eGFR (max 1000 mg/day at 30–44; stop <30) • Before pregnancy: both classes must be stopped (Insulin instead)
Q3. What will you do to her Levothyroxine?	• Under-replaced: TSH high with low T4 • First check: adherence and administration (empty stomach, 30–60 min before breakfast; separate from calcium, iron, antacids and PPI by 4 hours) • Then increase by 12.5–25 mcg (to 112–125 mcg) • Recheck TSH in 6–8 weeks
Q4. The patient got pregnant: what will you do for her DM medications?	• Switch to Insulin (basal-bolus); Metformin may continue as an adjunct • Stop: GLP-1 RA, SGLT2 inhibitors, ACE inhibitor/ARB, statin • Targets: A1C <6% if safe; fasting 70–95, 1-hour postprandial <140, 2-hour <120 mg/dL • Add low-dose Aspirin 81–162 mg from 12 weeks (preeclampsia risk) • Monitor: SMBG or CGM, A1C monthly, kidney function and BP, retina exam
Q5. What will you do for her Levothyroxine in pregnancy?	• Increase the dose by about 25–30% as soon as pregnancy is confirmed (two extra tablets per week) • TSH target: <2.5 mIU/L • Monitor: TSH every 4 weeks until week 20, then once around week 30 • After delivery: return to the pre-pregnancy dose
Q6. Screen shows post-meal readings and Insulin doses: all high except post-dinner. What to do with her Insulin?	• Increase pre-meal (bolus) doses for breakfast and lunch by 10–20% (1–2 units) • Keep the dinner dose unchanged • Check fasting readings for basal adequacy • Reassess every 3 days; watch for hypoglycemia
Q7. She delivered and developed cardiomyopathy: which agents will you avoid?	• Thiazolidinediones (Pioglitazone): fluid retention • Saxagliptin and Alogliptin: higher HF hospitalization • Avoid NSAIDs and non-DHP CCBs • Favor: SGLT2 inhibitors (HF benefit) and GLP-1 RA
Q8. Her renal function deteriorated and she started HD: what will you do with her DM medications?	• Stop Metformin (contraindicated eGFR <30) • Insulin: reduce the dose by 25–50% (lower clearance); main agent • Linagliptin (no adjustment) or Sitagliptin 25 mg • GLP-1 RA: Dulaglutide, Liraglutide or Semaglutide with caution (GI losses) • Avoid: Glibenclamide, SGLT2 inhibitors for glucose lowering • Monitor: glucose logs or CGM (A1C is unreliable in HD), hypoglycemia; A1C target about 7.5–8%
Also be ready for (exam add-ons)	• Goals: A1C <7%, BP <130/80, prevent micro- and macrovascular complications, avoid hypoglycemia, kidney protection

Question	Answer
	• Non-pharm: weight loss 5–10%, DASH diet, sodium <1.5–2.3 g, 150 min/week exercise with resistance training, limit alcohol, stop smoking, foot and eye care • First-line: Metformin 500 mg with the largest meal then 1000 mg BID; HTN: ACE inhibitor/ARB + CCB or thiazide (Lisinopril 10 mg + Amlodipine 5 mg); add SGLT2 inhibitor or GLP-1 RA if ASCVD, HF or CKD • Pregnancy: stop ACE inhibitor/ARB, SGLT2 inhibitor, statins; use Insulin, Labetalol, Nifedipine or Methyldopa; low-dose Aspirin from 12 weeks • Renal/dialysis: Metformin eGFR 30–44: max 1000 mg/day, <30: stop; ACE inhibitor: SCr rise ≤30% acceptable; Linagliptin 5 mg needs no adjustment; Insulin dose reduced • Liver: avoid Metformin in decompensated cirrhosis; Pioglitazone and sulfonylurea caution • HF: avoid Pioglitazone and Saxagliptin/Alogliptin; prefer SGLT2 inhibitor • Counseling: Metformin with food, GI effects fade, B12 yearly, hold with contrast or acute illness; ACE inhibitor cough; Amlodipine edema; hypoglycemia not usual with Metformin • Nephropathy: ACR ≥30 mg/g: maximum-tolerated ACE inhibitor/ARB + SGLT2 inhibitor (eGFR ≥20) + Finerenone 10–20 mg (K ≤5, eGFR ≥25); GLP-1 RA also helps; BP <130/80 • Thyroid (Levothyroxine): take on an empty stomach 30–60 min before breakfast; separate from calcium, iron, antacids, PPI by 4 h; adjust by 12.5–25 mcg and recheck TSH in 6–8 weeks; pregnancy: increase dose 25–30% when pregnant

2024

Case 1: T2DM with Hypothyroidism and Albuminuria (Home Glucose Log) ★ Repeated case, also in 2022–2023

Case details: Diabetic patient with fatigue. A1C 10.1%, fasting glucose high, TSH high, T4 low, albumin:creatinine ratio 366 mg/g. Metformin 1 g BID and Levothyroxine 100 mcg daily.

Question	Answer
Q1. What will you add to his diabetes regimen?	• A1C >10% with symptoms: **basal Insulin Glargine 10 units daily** (0.1–0.2 unit/kg) + **SGLT2 inhibitor** (Empagliflozin 10 mg or Dapagliflozin 10 mg) for albuminuria and cardiovascular benefit • A GLP-1 RA (Semaglutide 0.25 mg weekly) is an alternative or addition (ADA: weight and organ benefit)
Q2. Non-pharmacological treatment?	• Medical nutrition therapy, 150 min/week activity with resistance training, weight loss 5–10%, sodium <2300 mg/day, smoking cessation, DSMES, foot and eye care
Q3. Classification and severity of the albuminuria?	• ACR 366 mg/g = **A3 (severely increased, >300)**; with eGFR it gives the KDIGO risk category (A3 = high to very high risk)
Q4. Which medicines have cardiovascular benefit?	• GLP-1 RA (Semaglutide, Liraglutide, Dulaglutide), SGLT2 inhibitors (Empagliflozin, Dapagliflozin, Canagliflozin), Finerenone (CKD + T2D), statin, ACE inhibitor/ARB
Q5. Home readings: fasting, post-breakfast and post-lunch high; post-dinner in range. Glargine 22 units, Aspart 6 units TID: how will you adjust?	• **Increase Glargine by 2 units** (fasting high) and **increase pre-breakfast and pre-lunch Aspart by 2 units each**; keep the dinner dose • Reassess in 3 days; reduce by 10–20% for any glucose <70 mg/dL
Q6. How will you treat the albuminuria?	• ACE inhibitor/ARB at the maximum tolerated dose (SCr and K at 1–2 weeks), SGLT2 inhibitor (eGFR ≥20), Finerenone 10–20 mg if K ≤5 and eGFR ≥25, GLP-1 RA, BP <130/80, sodium restriction; repeat ACR in 3–6 months
Q7. Does the Levothyroxine dose need adjustment?	• TSH high with low T4: **under-replaced**; check adherence, administration (empty stomach, 4 h from calcium/iron/PPI) and interacting drugs • Increase by 12.5–25 mcg (to 112–125 mcg) and recheck TSH in 6–8 weeks
Also be ready for (exam add-ons)	• Goals: A1C <7%, BP <130/80, prevent micro- and macrovascular complications, avoid hypoglycemia, kidney protection • Non-pharm: weight loss 5–10%, DASH diet, sodium <1.5–2.3 g, 150 min/week exercise with resistance training, limit alcohol, stop smoking, foot and eye care • First-line: Metformin 500 mg with the largest meal then 1000 mg BID; HTN: ACE inhibitor/ARB + CCB or thiazide (Lisinopril 10 mg + Amlodipine 5 mg); add SGLT2 inhibitor or GLP-1 RA if ASCVD, HF or CKD • Pregnancy: stop ACE inhibitor/ARB, SGLT2 inhibitor, statins; use Insulin, Labetalol, Nifedipine or Methyldopa; low-dose Aspirin from 12 weeks • Renal/dialysis: Metformin eGFR 30–44: max 1000 mg/day, <30: stop; ACE inhibitor: SCr rise ≤30% acceptable; Linagliptin 5 mg needs no adjustment; Insulin dose reduced • Liver: avoid Metformin in decompensated cirrhosis; Pioglitazone and sulfonylurea caution • HF: avoid Pioglitazone and Saxagliptin/Alogliptin; prefer SGLT2 inhibitor • Counseling: Metformin with food, GI effects fade, B12 yearly, hold with contrast or acute illness; ACE inhibitor cough; Amlodipine edema; hypoglycemia not usual with Metformin • Nephropathy: ACR ≥30 mg/g: maximum-tolerated ACE inhibitor/ARB + SGLT2 inhibitor (eGFR ≥20) + Finerenone 10–20 mg (K ≤5, eGFR ≥25); GLP-1 RA also helps; BP <130/80 • Thyroid (Levothyroxine): take on an empty stomach 30–60 min before breakfast; separate from calcium, iron, antacids, PPI by 4 h; adjust by 12.5–25 mcg and recheck TSH in 6–8 weeks; pregnancy: increase dose 25–30% when pregnant

2022–2023

Case 1: New T2DM with Hypertension ★ Repeated case, also in 2020

Case details: 60-year-old man (70 kg), T2DM (A1C 7%) and HTN (BP 150/95). Lifestyle modification failed.

Question	Answer
Q1. How would you manage his HTN?	• Stage 2 (≥140/90): start two first-line agents from different classes, ideally as a single pill ◦ ACE inhibitor or ARB (diabetes) plus a DHP CCB or thiazide ◦ Example: Lisinopril 10 mg + Amlodipine 5 mg daily • Lifestyle: weight loss, DASH diet, sodium <1500–2300 mg/day, exercise, limit alcohol • Monitor: BP at 4 weeks; SCr and K at 1–2 weeks
Q2. What is your target BP according to recent ADA guidelines?	• <130/80 mm Hg
Q3. How would you manage his DM and advise him to take it (dose and administration)?	• A1C 7% is mildly above goal: start Metformin 500 mg PO once daily with the largest meal • Titrate by 500 mg every 1–2 weeks to 1000 mg BID as tolerated • Counsel: take with food, GI effects usually fade, report vomiting or dehydration • Monitor: A1C every 3 months, eGFR yearly, B12 every 2–3 years • Non-pharm: 150 minutes/week activity, weight loss 5–10%
Q4. Kidney function deteriorated to CrCl <30. What would you do to Metformin?	• Stop Metformin (contraindicated at eGFR <30; do not start at 30–44) • Alternatives: DPP-4 inhibitor Linagliptin 5 mg, GLP-1 RA, SGLT2 inhibitor for kidney benefit, Insulin
Q5. Lisinopril 10 mg started: how would you manage a decline in renal function (no baseline)?	• An SCr rise up to 30% is acceptable; continue • >30%: look for volume depletion, NSAIDs, renal artery stenosis; reduce or hold the dose • K >5.5: reduce or stop • Monitor: SCr and K 1–2 weeks after start and each dose change
Q6. Non-pharmacological approaches to control his BP?	• Weight loss (about 1 mm Hg per kg) • DASH diet • Sodium <1500–2300 mg/day • Regular aerobic and resistance exercise • Limit alcohol; stop smoking
Also be ready for (exam add-ons)	• Goals: A1C <7%, BP <130/80, prevent micro- and macrovascular complications, avoid hypoglycemia, kidney protection • Non-pharm: weight loss 5–10%, DASH diet, sodium <1.5–2.3 g, 150 min/week exercise with resistance training, limit alcohol, stop smoking, foot and eye care • First-line: Metformin 500 mg with the largest meal then 1000 mg BID; HTN: ACE inhibitor/ARB + CCB or thiazide (Lisinopril 10 mg + Amlodipine 5 mg); add SGLT2 inhibitor or GLP-1 RA if ASCVD, HF or CKD • Pregnancy: stop ACE inhibitor/ARB, SGLT2 inhibitor, statins; use Insulin, Labetalol, Nifedipine or Methyldopa; low-dose Aspirin from 12 weeks • Renal/dialysis: Metformin eGFR 30–44: max 1000 mg/day, <30: stop; ACE inhibitor: SCr rise ≤30% acceptable; Linagliptin 5 mg needs no adjustment; Insulin dose reduced • Liver: avoid Metformin in decompensated cirrhosis; Pioglitazone and sulfonylurea caution • HF: avoid Pioglitazone and Saxagliptin/Alogliptin; prefer SGLT2 inhibitor • Counseling: Metformin with food, GI effects fade, B12 yearly, hold with contrast or acute illness; ACE inhibitor cough; Amlodipine edema; hypoglycemia not usual with Metformin

Case 2: T2DM with Hypothyroidism and Nephropathy

Case details: Polydipsia and thirst. T2DM, low T4, high TSH, high BP. A1C 10.4%. On Metformin 1 g BID, Levothyroxine 100 mcg, Amlodipine 10 mg, antacids PRN. Albumin:creatinine ratio 312.

Question	Answer
Q1. What confirms T2DM (signs, symptoms and labs)?	• Symptoms: polydipsia, polyuria, thirst, weight loss • A1C ≥6.5% • Fasting glucose ≥126 mg/dL (7.0 mmol/L) • 2-hour OGTT or random glucose ≥200 mg/dL (11.1 mmol/L) with symptoms • Confirm with a repeat test unless clearly symptomatic
Q2. What is the best management, and the doses?	• A1C >10%: Insulin is the reasonable choice ◦ Basal Insulin (Glargine) 10 units daily or 0.1–0.2 unit/kg; titrate by 2 units every 3 days to fasting 80–130 mg/dL • Continue Metformin (check eGFR); add SGLT2 inhibitor or GLP-1 RA for kidney benefit • Kidney: ACE inhibitor or ARB at maximum tolerated dose • Monitor: SMBG, hypoglycemia, A1C every 3 months, SCr, K, urine ACR
Q3. If HF develops, which anti-diabetic medication should be avoided?	• Thiazolidinediones (Pioglitazone) • Saxagliptin and Alogliptin • Prefer: SGLT2 inhibitors

Question	Answer
Q4. What does the Albumin:creatinine ratio mean here, and the classification?	• Diabetic kidney disease with albuminuria • ACR 312 mg/g = A3 (severely increased; >300) • Staging: A1 <30, A2 30–300, A3 >300 • Treat: ACE inhibitor/ARB, SGLT2 inhibitor, Finerenone, BP and glucose control
Q5. What to do with the Levothyroxine dose? Counseling points?	• First assess adherence and administration before increasing the dose • Then increase to 125 mcg daily • Counsel: take in the morning on an empty stomach, 30–60 minutes before food • Separate from antacids, calcium and iron by 4 hours • Monitor: TSH 6–8 weeks after each change
Also be ready for (exam add-ons)	• Goals: A1C <7%, BP <130/80, prevent micro- and macrovascular complications, avoid hypoglycemia, kidney protection • Non-pharm: weight loss 5–10%, DASH diet, sodium <1.5–2.3 g, 150 min/week exercise with resistance training, limit alcohol, stop smoking, foot and eye care • First-line: Metformin 500 mg with the largest meal then 1000 mg BID; HTN: ACE inhibitor/ARB + CCB or thiazide (Lisinopril 10 mg + Amlodipine 5 mg); add SGLT2 inhibitor or GLP-1 RA if ASCVD, HF or CKD • Pregnancy: stop ACE inhibitor/ARB, SGLT2 inhibitor, statins; use Insulin, Labetalol, Nifedipine or Methyldopa; low-dose Aspirin from 12 weeks • Renal/dialysis: Metformin eGFR 30–44: max 1000 mg/day, <30: stop; ACE inhibitor: SCr rise ≤30% acceptable; Linagliptin 5 mg needs no adjustment; Insulin dose reduced • Liver: avoid Metformin in decompensated cirrhosis; Pioglitazone and sulfonylurea caution • HF: avoid Pioglitazone and Saxagliptin/Alogliptin; prefer SGLT2 inhibitor • Counseling: Metformin with food, GI effects fade, B12 yearly, hold with contrast or acute illness; ACE inhibitor cough; Amlodipine edema; hypoglycemia not usual with Metformin • Nephropathy: ACR ≥30 mg/g: maximum-tolerated ACE inhibitor/ARB + SGLT2 inhibitor (eGFR ≥20) + Finerenone 10–20 mg (K ≤5, eGFR ≥25); GLP-1 RA also helps; BP <130/80 • Thyroid (Levothyroxine): take on an empty stomach 30–60 min before breakfast; separate from calcium, iron, antacids, PPI by 4 h; adjust by 12.5–25 mcg and recheck TSH in 6–8 weeks; pregnancy: increase dose 25–30% when pregnant

Case 3: Diabetes with Multiple Comorbidities

Case details: Diabetic, obese patient with HTN, dyslipidemia and osteoporosis with a prior fracture.

Question	Answer
Q1. What are the non-pharmacological measures for diabetes?	• Weight reduction (5–10%) • Physical activity: at least 150 minutes/week plus resistance training • Medical nutrition therapy and carbohydrate awareness • Smoking cessation, limit alcohol • Sleep and stress management
Q2. Tingling in his feet: what is it, how is it diagnosed, and how is it treated?	• Diabetic peripheral neuropathy • Diagnosis: 10-g monofilament, vibration (128 Hz tuning fork), pinprick, ankle reflexes • First-line: Duloxetine 30 → 60 mg daily; Pregabalin 75 → 150 mg BID; Gabapentin 300 mg TID titrated • Alternatives: Amitriptyline 10–25 mg at bedtime (avoid in elderly or cardiac disease), sodium channel blockers • Also: optimize glucose, B12 check on Metformin, daily foot care
Q3. What are the side effects of Insulin?	• Hypoglycemia • Weight gain • Injection-site reactions, lipohypertrophy • Hypokalemia (IV use)
Q4. Counseling of Insulin?	• Storage: unopened in the refrigerator; opened pen at room temperature, discard after 28–30 days • Technique: wash hands, check the label and clarity, prime the pen, new needle each time • Sites: abdomen, thigh, upper arm, buttocks; rotate sites, stay 2 inches from the umbilicus; avoid lumps • Hypoglycemia: recognize symptoms, treat with 15 g fast carbohydrate and recheck in 15 minutes • Mealtime Insulin: inject 0–15 minutes before meals; skip it if the meal is skipped • Monitoring: SMBG, bring logs, sharps disposal
Q5. What will you give the patient for osteoporosis? With counseling.	• Alendronate 70 mg PO once weekly (CrCl >35) • Counsel: take first thing in the morning with 8 oz plain water, 30 minutes before food or other drugs, remain upright 30 minutes • Calcium 1200 mg/day (diet plus supplement) and vitamin D 800–1000 IU/day • Report: chest pain, difficulty swallowing, thigh pain, jaw pain • Monitor: DEXA every 1–2 years, calcium, SCr; reassess after 5 years
Q6. Another patient with type 1 DM: calculate the initial Insulin regimen.	• Total daily dose 0.4–0.6 unit/kg/day (example 0.5) • 50% basal (Glargine once daily) and 50% bolus divided before the three meals • Example (70 kg): TDD 35 units → Glargine 17–18 units + about 6 units before each meal • Titrate by SMBG every 3 days

Question	Answer
Q7. Metformin 2 g/day, Gliclazide, Empagliflozin with eGFR 35: what will you adjust? Side effects of Dapagliflozin?	• Metformin: reduce to 500 mg BID (max 1000 mg/day at eGFR 30–44) • Empagliflozin: continue for kidney and HF benefit (glucose effect is smaller) • Gliclazide: risk of hypoglycemia, reduce or stop if lows • SGLT2 inhibitor side effects: genital mycotic infections, UTI, euglycemic DKA, volume depletion and hypotension, AKI, Fournier gangrene • Monitor: eGFR, volume status, ketones if ill, glucose
Also be ready for (exam add-ons)	• Goals: A1C individualized, fracture prevention, neuropathic pain control, BP and lipid control • Non-pharm: weight loss, weight-bearing exercise, fall prevention, calcium/Vitamin D intake, foot care, stop smoking/alcohol • First-line: Metformin ± SGLT2 inhibitor or GLP-1 RA; neuropathy: Duloxetine 30 → 60 mg, Pregabalin 75 → 150 mg BID or Gabapentin; osteoporosis: Alendronate 70 mg weekly (CrCl >35) with Calcium 1200 mg and Vitamin D 800–1000 units • Pregnancy: stop statin, ACE inhibitor/ARB, bisphosphonate (long retention), Duloxetine (caution), Pregabalin; use Insulin, Labetalol, Nifedipine; neuropathic pain: Amitriptyline or Gabapentin only if needed • Renal/dialysis: Alendronate avoid CrCl <35 (use Denosumab with Ca/Vitamin D monitoring); Pregabalin 25–150 mg/day by CrCl; Gabapentin by CrCl; Duloxetine avoid <30; Metformin eGFR rules • Liver: Duloxetine avoid in hepatic impairment/alcohol; statin caution • GI (PUD/GERD): Alendronate causes esophagitis (avoid with active GERD/dysphagia; use Zoledronic acid IV); NSAID avoidance; PPI lowers calcium carbonate absorption (use citrate) • Counseling: Alendronate on an empty stomach with 8 oz water, upright 30 min; Duloxetine nausea and BP, do not stop abruptly; Pregabalin/Gabapentin dizziness; Insulin counseling as above

Case 4: Thyroid Emergencies (Thyroid Storm and Myxedema Coma) ★ Repeated case, also in 2021, 2020, 2018–2019

Case details: Hyperthyroid crisis and hypothyroid crisis.

Question	Answer
Q1. What is the diagnosis?	• Thyroid storm: fever, tachycardia, agitation, GI and hepatic dysfunction; Burch-Wartofsky score ≥45 • Myxedema coma: hypothermia, bradycardia, hypoventilation, altered consciousness, hyponatremia • Both: a clinical diagnosis; look for a precipitant (infection, surgery, stopped drugs)
Q2. Treatment and doses?	• Thyroid storm: ◦ Propylthiouracil 500–1000 mg load then 250 mg q4h (or Methimazole 60–80 mg/day) ◦ Iodine (Lugol 8 drops or Potassium iodide 5 drops q6h) at least 1 hour after the thionamide ◦ Propranolol 60–80 mg PO q4–6h (or Esmolol infusion) ◦ Hydrocortisone 100 mg IV q8h ◦ Cool with Acetaminophen (avoid Aspirin); Cholestyramine 4 g q6h if needed; treat the precipitant • Myxedema coma: ◦ Levothyroxine 200–400 mcg IV load, then 1.6 mcg/kg/day (75% IV); use the low end in the elderly and cardiac patients ◦ ± Liothyronine 5–20 mcg IV load then 2.5–10 mcg q8h ◦ Hydrocortisone 100 mg IV q8h before thyroid hormone ◦ Passive warming, ventilatory support, isotonic fluids, treat infection • Chronic therapy for comparison: Levothyroxine 1.6 mcg/kg/day (25 mcg in cardiac patients); Methimazole 10–30 mg daily • Monitor: ICU, ECG, temperature, free T4/T3, TSH, glucose, electrolytes, LFTs
Also be ready for (exam add-ons)	• Goals: stabilize vital signs, block hormone synthesis/release/action (storm) or replace hormone safely (myxedema), treat the precipitant, prevent relapse • Non-pharm: ICU monitoring, cooling (storm) or passive rewarming (myxedema), IV fluids, ventilation, treat infection, avoid Aspirin in storm, avoid sedatives in myxedema • First-line: storm: Propylthiouracil 500–1000 mg load then 250 mg q4h (or Methimazole 60–80 mg/day) → Lugol 8 drops or Potassium iodide 5 drops q6h ≥1 h later → Propranolol 60–80 mg q4–6h → Hydrocortisone 100 mg IV q8h; myxedema: Levothyroxine 200–400 mcg IV load then 1.6 mcg/kg/day (75% IV), ± Liothyronine, Hydrocortisone 100 mg IV q8h first • Pregnancy: Propylthiouracil in the first trimester then Methimazole; storm management otherwise the same (Propranolol short-term); Levothyroxine dose up 25–30%; avoid Radioiodine • Renal/dialysis: Propylthiouracil/Methimazole no adjustment; Levothyroxine dose by weight; Liothyronine caution • Liver: Propylthiouracil carries liver-failure boxed warning (avoid except first trimester/storm); Methimazole cholestasis; Levothyroxine reduce in cirrhosis • HF/cardiac: myxedema: low-dose Levothyroxine (25–50 mcg) and cardiac monitoring; storm: Propranolol avoided in decompensated HF (use Esmolol with care) • Counseling: Levothyroxine on an empty stomach, lifelong; antithyroid drugs: fever or sore throat means stop and call (agranulocytosis), jaundice, rash; adherence prevents relapse

Case 5: DKA (Staging and Step-by-Step Treatment) ★ Repeated case, also in 2025, 2021, 2018–2019

Case details: DKA case.

Question	Answer
Q1. Which stage?	• Mild: pH 7.25–7.30, bicarbonate 15–18, alert • Moderate: pH 7.00–7.24, bicarbonate 10–15, drowsy • Severe: pH <7.00, bicarbonate <10, stupor or coma
Q2. Treatment step by step, based on electrolytes and glucose?	• Step 1: 0.9% NaCl 15–20 mL/kg in the first hour • Step 2 (check K first): ◦ K <3.3: hold Insulin, give 20–30 mEq/h potassium until >3.3 ◦ K 3.3–5.2: add 20–30 mEq potassium per liter of fluid ◦ K >5.2: no potassium; recheck every 2 hours • Step 3: Insulin 0.1 unit/kg IV bolus, then 0.1 unit/kg/h; fall of 50–75 mg/dL/h • Step 4: at glucose <200 add dextrose 5% and reduce Insulin to 0.02–0.05 unit/kg/h • Step 5: bicarbonate only if pH <7.0 • Resolution: glucose <200 mg/dL and two of: anion gap ≤12, bicarbonate ≥15, pH >7.3 • Then transition to SC Insulin with a 1–2 hour overlap
Also be ready for (exam add-ons)	• Goals: correct dehydration, hyperglycemia, ketoacidosis and electrolytes, treat the precipitant, then transition to SC Insulin and prevent recurrence • Non-pharm: ICU/step-down monitoring, NPO initially, strict intake/output, treat infection source, sick-day and ketone education, assess adherence barriers (cost, fear, technique) • First-line: 0.9% NaCl 15–20 mL/kg first hour; regular Insulin 0.1 unit/kg IV bolus then 0.1 unit/kg/h (only if K ≥3.3); K 20–30 mEq/L once K <5.2; Dextrose 5% when glucose <200; Bicarbonate only if pH <7.0; resolve: glucose <200 + 2 of HCO3 ≥15, pH >7.3, AG ≤12 • Pregnancy: DKA occurs at lower glucose levels (euglycemic) and is an emergency for the fetus: same protocol with fetal monitoring, left lateral position, K and phosphate care; Insulin is the only agent; SGLT2 inhibitors and GLP-1 RA stopped • Renal/dialysis: reduce Insulin needs (lower clearance), watch K and phosphate, careful fluids in oliguria; Metformin hold; Glibenclamide avoid • Liver: hypoglycemia risk (gluconeogenesis impaired); lactate interpretation • HF/CKD: fluids 250–500 mL/h guided by volume status; monitor for pulmonary edema • Counseling: never stop basal Insulin, sick-day rules, check ketones when glucose >250 or ill, SMBG 4×/day, hypoglycemia 15-15 rule, return precautions, SGLT2 inhibitors can cause euglycemic DKA • Monitoring: glucose hourly; BMP/venous gas every 2–4 h; K every 2 h at first; β-hydroxybutyrate; ECG; urine output

Case 6: Insulin Initiation and Counseling ★ Repeated case, also in 2022–2023, 2021, 2018–2019, 2020

Case details: Patient with DM who will start Insulin.

Question	Answer
Q1. Calculate the Insulin dose and counsel how to use Insulin.	• Type 1: 0.4–0.6 unit/kg/day; half basal, half bolus divided before meals • Type 2: basal Insulin 0.1–0.2 unit/kg/day (or 10 units) daily; add 4–6 units of prandial Insulin to the largest meal if needed • Counseling: wash hands, check the label, clarity and expiry; rotate sites; do not inject into lumps, bruised or scarred skin • Do not mix Insulin glargine with other insulins; do not use cloudy or discolored solution (except NPH) • Pens: prime, hold the needle in for 10 seconds, remove the needle after each use; do not count clicks to prepare a dose • Missed dose: take it when remembered unless it is close to the next dose; never double up • Storage: unopened in the refrigerator, do not freeze; opened pen up to 28–30 days at room temperature, away from heat and light • Hypoglycemia: symptoms and the 15-15 rule; sharps disposal; follow the diet and exercise plan
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals,

Question	Answer
	skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

Case 7: DM (Insulin Calculations) ★ Repeated case, also in 2022–2023, 2021, 2018–2019, 2020

Case details: DM case with Insulin calculations.

Question	Answer
Q1. Insulin calculations?	• Total daily dose: 0.4–0.6 unit/kg/day for T1DM; 0.5–0.8 unit/kg/day for a new Insulin requirement • Basal: 50% of TDD; bolus: 50% divided before 3 meals • Insulin-to-carbohydrate ratio: 500 ÷ TDD (g per unit) • Correction factor: 1800 ÷ TDD (mg/dL per unit) • Example (TDD 40): ICR 1:12 g; correction 1 unit per 45 mg/dL above target
Q2. Non-pharmacological management?	• Medical nutrition therapy • Physical activity ≥150 min/week • Weight loss 5–10% if overweight • Self-monitoring and education • Smoking cessation
Q3. Lifestyle modification?	• Consistent carbohydrate intake, fiber and whole grains • Limit sugar-sweetened beverages and saturated fat • Regular meals, avoid skipping meals when on Insulin • Sleep 7–8 hours, stress management • Foot, eye and dental care
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

Case 8: Type 1 DM (Insulin Regimen) ★ Repeated case, also in 2022–2023, 2021, 2018–2019, 2020

Case details: Type 1 DM, A1C 12%, FBG 280 mg/dL, weight 60 kg, no risk of hypoglycemia.

Question	Answer
Q1. Construct the Insulin regimen.	• TDD: 0.5–0.6 unit/kg × 60 kg = 30–36 units • Basal: Glargine about 15–18 units once daily at the same time • Bolus: Aspart or Lispro 5–6 units before each meal (0–15 minutes) • Correction factor: 1800 ÷ 36 = 1 unit per 50 mg/dL above 150 • Titrate: basal by 2 units every 3 days to fasting 80–130 mg/dL • Monitor: SMBG or CGM, A1C every 3 months
Q2. Side effects: Insulin and Dapagliflozin?	• Insulin: hypoglycemia, weight gain, lipohypertrophy, injection-site reactions • Dapagliflozin: genital infections, UTI, euglycemic DKA (especially in T1DM, not indicated), volume depletion, AKI
Q3. Counsel for Insulin use.	• Technique: new needle each time, rotate sites, inject into subcutaneous fat • Timing: basal at the same time daily; bolus 0–15 minutes before meals • Hypoglycemia: treat with 15 g carbohydrate, recheck in 15 minutes • Sick days: never stop basal Insulin; check ketones • Storage and sharps disposal • Monitor: SMBG before meals and at bedtime, carry glucose tablets

Question	Answer
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

2021

Case 1: Hypothyroidism and Hyperthyroidism (Graves) ★ Repeated case, also in 2022–2023, 2020, 2018–2019

Case details: A case of hypothyroidism and a case of hyperthyroidism (Graves disease).

Question	Answer
Q1. Hypothyroidism: indication, regimen and counseling?	• Indication: raised TSH with low free T4 (overt), or symptomatic subclinical disease with high TSH • Levothyroxine 1.6 mcg/kg/day; start 25–50 mcg/day in the elderly or with coronary disease, then titrate every 6–8 weeks • Counsel: empty stomach 30–60 minutes before breakfast, same time daily; separate from calcium, iron, antacids and PPI by 4 hours • Interactions: Rifampin, Phenytoin, Carbamazepine, estrogen raise the needed dose • Response: symptoms improve in 2–3 weeks; TSH normalizes in 6–8 weeks • Monitor: TSH every 6–8 weeks until stable, then yearly
Q2. Hyperthyroidism (Graves): how to manage, and for how long?	• Antithyroid drug: Methimazole 10–30 mg daily (Propylthiouracil 50–150 mg q8h in the first trimester or thyroid storm) • Symptoms: Propranolol 10–40 mg q6–8h • Duration: 12–18 months, then stop if TSH and TRAb are normal; otherwise definitive therapy (radioactive iodine or thyroidectomy) • Monitor: free T4/T3 and TSH every 4–6 weeks at first; baseline CBC with differential and LFTs • Warn: fever, sore throat, jaundice or dark urine → stop and call (agranulocytosis, hepatotoxicity)
Also be ready for (exam add-ons)	• Goals: normalize TSH and free T4 (hypothyroidism) or free T4/T3 (hyperthyroidism), relieve symptoms, prevent complications and relapse • Non-pharm: iodine intake awareness, smoking cessation (Graves ophthalmopathy; moderate to severe active orbitopathy: Teprotumumab, an anti-IGF-1R monoclonal antibody, or IV Methylprednisolone), sun/eye protection, avoid biotin before tests, adherence • First-line: hypothyroidism: Levothyroxine 1.6 mcg/kg/day (start 25–50 mcg in elderly or CAD), recheck TSH 6–8 weeks; hyperthyroidism: Methimazole 10–30 mg daily (12–18 months) + Propranolol 10–40 mg q6–8h; Radioiodine or surgery for definitive therapy • Pregnancy: hypothyroid: increase Levothyroxine 25–30% at once (two extra tablets/week), TSH <2.5, check every 4 weeks until week 20; hyperthyroid: Propylthiouracil in the first trimester, Methimazole after; avoid Radioiodine; postpartum return to pre-pregnancy dose • Renal/dialysis: Levothyroxine no adjustment (monitor TSH); Methimazole no adjustment • Liver: Propylthiouracil hepatotoxic; Methimazole cholestasis; baseline LFTs • HF/CAD: Levothyroxine start low (12.5–25 mcg); Propranolol caution in decompensated HF • GI: PPIs, calcium, iron, antacids, Sucralfate and cholestyramine reduce Levothyroxine absorption (separate by 4 h) • Counseling: timing, lifelong therapy, symptoms improve in 2–3 weeks and TSH in 6–8 weeks; Methimazole: agranulocytosis, rash, teratogenic (aplasia cutis); report palpitations or chest pain

Case 2: DKA (Multiple Scenarios) ★ Repeated case, also in 2025, 2022–2023, 2018–2019

Case details: DKA with multiple scenarios.

Question	Answer
Q1. Provide full management with targets.	• Fluids: 0.9% NaCl 15–20 mL/kg first hour, then 250–500 mL/h; add dextrose 5% when glucose <200 mg/dL • Potassium: <3.3 hold Insulin and replace; 3.3–5.2 add 20–30 mEq/L; >5.2 none • Insulin: 0.1 unit/kg IV bolus then 0.1 unit/kg/h; target fall 50–75 mg/dL/h; at glucose 200 reduce to 0.02– 0.05 unit/kg/h • Bicarbonate only if pH <7.0 • Targets: glucose 150–200 until resolved; K 4–5 mEq/L • Resolution: glucose <200 plus two of bicarbonate ≥15, pH >7.3, anion gap ≤12 • Transition: SC Insulin 0.5–0.8 unit/kg/day with 1–2 hour overlap • Monitor: glucose hourly, BMP and gas every 2–4 h, ketones, ECG, urine output
Also be ready for (exam add-ons)	• Goals: correct dehydration, hyperglycemia, ketoacidosis and electrolytes, treat the precipitant, then transition to SC Insulin and prevent recurrence • Non-pharm: ICU/step-down monitoring, NPO initially, strict intake/output, treat infection source, sick-day and ketone education, assess adherence barriers (cost, fear, technique) • First-line: 0.9% NaCl 15–20 mL/kg first hour; regular Insulin 0.1 unit/kg IV bolus then 0.1 unit/kg/h (only if K ≥3.3); K 20–30 mEq/L once K <5.2; Dextrose 5% when glucose <200; Bicarbonate only if pH <7.0; resolve: glucose <200 + 2 of HCO3 ≥15, pH >7.3, AG ≤12 • Pregnancy: DKA occurs at lower glucose levels (euglycemic) and is an emergency for the fetus: same protocol with fetal monitoring, left lateral position, K and phosphate care; Insulin is the only agent; SGLT2 inhibitors and GLP-1 RA stopped • Renal/dialysis: reduce Insulin needs (lower clearance), watch K and phosphate, careful fluids in oliguria; Metformin hold; Glibenclamide avoid • Liver: hypoglycemia risk (gluconeogenesis impaired); lactate interpretation • HF/CKD: fluids 250–500 mL/h guided by volume status; monitor for pulmonary edema • Counseling: never stop basal Insulin, sick-day rules, check ketones when glucose >250 or ill, SMBG 4×/day, hypoglycemia 15-15 rule, return precautions, SGLT2 inhibitors can cause euglycemic DKA • Monitoring: glucose hourly; BMP/venous gas every 2–4 h; K every 2 h at first; β-hydroxybutyrate; ECG; urine output

Case 3: Insulin (TDD from NPH) ★ Repeated case, also in 2022–2023, 2018–2019, 2020

Case details: Insulin conversion and counseling.

Question	Answer
Q1. Calculate the TDD (basal/bolus) from the NPH dose (just sum the units).	• TDD: add all NPH (and any regular) units given in 24 hours • Basal-bolus conversion: basal 50% (Glargine once daily), bolus 50% divided before meals • Example: NPH 20 units AM + 10 units PM = 30 units → Glargine 15 units + Aspart 5 units before each meal • If hypoglycemia risk: reduce the starting TDD by 10–20% • Counseling and monitoring: SMBG 4 times daily, adjust every 3 days
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

2020

Cases 1 and 3: DM with HTN, Dyslipidemia, Obesity and Sleep Problem ★ Repeated case, also in 2022–2023, 2020

Case details: 40-year-old with BMI >30, high BP, high A1C and a sleep problem. Metformin 500 mg only.

Question	Answer
Q1. What is the problem list? What are the diseases?	• Type 2 DM (uncontrolled) • Hypertension • Obesity (BMI >30) • Dyslipidemia • Sleep problem: screen for obstructive sleep apnea (STOP-BANG, sleep study)
Q2. What agents to add for BP? Which statin group?	• ACE inhibitor or ARB (diabetes): Lisinopril 10 mg daily; target <130/80 • Add Amlodipine 5 mg or a thiazide if still above goal • Statin: DM aged 40–75 → moderate intensity; high intensity (Atorvastatin 40–80 mg or Rosuvastatin 20–40 mg) with multiple risk factors • Monitor: SCr and K in 1–2 weeks, lipids in 4–12 weeks, BP in 4 weeks
Q3. Metformin 500 only: what to modify?	• Titrate Metformin to 1000 mg BID (500 mg weekly increments) • Add a GLP-1 RA or GIP/GLP-1 RA when A1C is still above goal (also treats obesity) ◦ Semaglutide 0.25 mg SC weekly → 0.5 → 1 mg; or Tirzepatide 2.5 mg weekly, +2.5 mg every 4 weeks • Do not combine a GLP-1 RA with a DPP-4 inhibitor (stop Sitagliptin)
Q4. LDL decreased after GLP-1 RA use: what is the explanation?	• Weight loss and better Insulin sensitivity • Lower hepatic VLDL production and less intestinal chylomicron secretion • Lower triglycerides, modest LDL effect
Q5. Patient after 3 months not improved: plan? Why did the patient improve (adherence, non-pharm)?	• Assess adherence, technique, diet, activity, side effects and cost • Intensify: titrate the GLP-1 RA, add SGLT2 inhibitor, or add basal Insulin • Sleep: treat OSA (CPAP); Tirzepatide is also approved for OSA with obesity • Non-pharm: weight loss 5–10%, activity ≥150 min/week, nutrition referral • Monitor: A1C every 3 months, weight, BP, lipids
Also be ready for (exam add-ons)	• Goals: A1C <7%, weight loss ≥5–10%, BP <130/80, LDL <70 (<55 with ASCVD), treat sleep apnea • Non-pharm: medical nutrition therapy, 150 min/week exercise, weight loss, CPAP for OSA, smoking cessation, sleep hygiene, alcohol limit • First-line: Metformin to 1000 mg BID; add GLP-1 RA (Semaglutide 0.25 → 0.5 → 1 mg weekly) or Tirzepatide (2.5 mg weekly, +2.5 mg every 4 weeks; also treats OSA); ACE inhibitor/ARB for BP; moderate–high-intensity statin • Pregnancy: stop GLP-1 RA/Tirzepatide ≥2 months before conception (limited data, may reduce OC efficacy); use Insulin; stop ACE inhibitor/ARB and statin; Labetalol or Nifedipine • Renal/dialysis: Metformin eGFR rules; GLP-1 RA no adjustment (watch dehydration); SGLT2 inhibitor for CKD • Liver: GLP-1 RA safe in MASLD; Pioglitazone for NASH caution (HF, fractures) • HF: avoid Pioglitazone and Saxagliptin; SGLT2 inhibitor beneficial • Counseling: GLP-1 RA nausea (small meals), pancreatitis and gallbladder symptoms, MTC/MEN2 contraindication, injection technique, dehydration; Metformin with food • Monitoring: A1C every 3 months, weight, BP, lipids, eGFR, B12, retinal exam (rapid A1C drop)

Case 2: DM (Detailed) ★ Repeated case, also in 2022–2023, 2020

Case details: DM patient, weight 100 kg, A1C 6.8%, BP above target.

Question	Answer
Q1. List and prioritize the problems.	• 1. Hypertension not at goal (highest CV risk) • 2. Obesity (100 kg) • 3. Dyslipidemia (statin indicated by diabetes) • 4. T2DM near goal (A1C 6.8%)
Q2. Which medications will you start?	• ACE inhibitor or ARB: Lisinopril 10 mg daily • Metformin 500 mg daily with meals, titrate to 1000 mg BID • Statin: Rosuvastatin 10–20 mg or Atorvastatin 20–40 mg • Monitor: SCr, K, lipids, A1C
Q3. Started on Lisinopril, Metformin and Rosuvastatin; A1C still not controlled after 3 months. What will you do?	• Check adherence and Metformin titration (up to 2000 mg/day) • Add GLP-1 RA, Tirzepatide or SGLT2 inhibitor (weight and CV benefit) • Basal Insulin if A1C remains high • Monitor: A1C every 3 months, weight, GI effects

Question	Answer
Q4. What other measures can help control his DM?	• Weight loss 5–10% • ≥150 minutes/week activity and resistance training • Medical nutrition therapy, SMBG or CGM • Annual eye, foot and kidney screening • Vaccines, smoking cessation
Also be ready for (exam add-ons)	• Goals: A1C <7%, weight loss ≥5–10%, BP <130/80, LDL <70 (<55 with ASCVD), treat sleep apnea • Non-pharm: medical nutrition therapy, 150 min/week exercise, weight loss, CPAP for OSA, smoking cessation, sleep hygiene, alcohol limit • First-line: Metformin to 1000 mg BID; add GLP-1 RA (Semaglutide 0.25 → 0.5 → 1 mg weekly) or Tirzepatide (2.5 mg weekly, +2.5 mg every 4 weeks; also treats OSA); ACE inhibitor/ARB for BP; moderate–high-intensity statin • Pregnancy: stop GLP-1 RA/Tirzepatide ≥2 months before conception (limited data, may reduce OC efficacy); use Insulin; stop ACE inhibitor/ARB and statin; Labetalol or Nifedipine • Renal/dialysis: Metformin eGFR rules; GLP-1 RA no adjustment (watch dehydration); SGLT2 inhibitor for CKD • Liver: GLP-1 RA safe in MASLD; Pioglitazone for NASH caution (HF, fractures) • HF: avoid Pioglitazone and Saxagliptin; SGLT2 inhibitor beneficial • Counseling: GLP-1 RA nausea (small meals), pancreatitis and gallbladder symptoms, MTC/MEN2 contraindication, injection technique, dehydration; Metformin with food • Monitoring: A1C every 3 months, weight, BP, lipids, eGFR, B12, retinal exam (rapid A1C drop)

Case 4: DM with CKD 3b, HF and Pancreatitis

Case details: 40-year-old with T2DM, HF (EF 40s), CKD 3b, chronic pancreatitis, on Insulin. Albuminuria. A1C 7.9%. Team asks about starting Metformin.

Question	Answer
Q1. Which class would you avoid in a patient with pancreatitis?	• GLP-1 receptor agonists • DPP-4 inhibitors
Q2. Which classes to avoid in a patient with HF?	• Thiazolidinediones (Pioglitazone) • Saxagliptin (and Alogliptin) from the DPP-4 class
Q3. Which class would you add to control his DM?	• SGLT2 inhibitor: Dapagliflozin 10 mg or Empagliflozin 10 mg daily (eGFR ≥20–25) • Metformin: do not start at eGFR 30–44
Q4. Why add what you chose?	• Diabetes + HF + CKD with albuminuria: SGLT2 inhibitors reduce HF hospitalization, CKD progression and CV death • Approved for diabetic kidney disease with proteinuria
Q5. What would you monitor with this new medication?	• Volume status: hypotension, falls (consider reducing the diuretic) • SCr: expect an initial eGFR dip, accept ≤30% • A1C, fasting and random glucose • Genital infection and UTI; euglycemic DKA (hold 3 days before surgery) • K, foot care
Also be ready for (exam add-ons)	• Goals: lower A1C safely, slow CKD progression, reduce HF hospitalization, avoid hypoglycemia and pancreatitis • Non-pharm: sodium restriction, weight/volume monitoring, exercise, protein 0.8 g/kg, alcohol avoidance (pancreatitis), stop smoking, vaccination • First-line: SGLT2 inhibitor (Dapagliflozin 10 mg or Empagliflozin 10 mg daily, eGFR ≥20–25) + Insulin; avoid GLP-1 RA/DPP-4 inhibitors (pancreatitis), Pioglitazone and Saxagliptin (HF); Metformin not started at eGFR 30–44 • Pregnancy: SGLT2 inhibitors and ACE inhibitor/ARB/MRA contraindicated; use Insulin; hydralazine + nitrate for HF • Renal/dialysis: SGLT2 inhibitor expect an initial eGFR dip ≤30%; hold on sick days and 3 days before surgery; Insulin dose reduction • Liver: pancreatitis can have biliary/alcohol causes; Insulin preferred when hepatic impairment • Counseling: genital hygiene, hydration, sick-day rules (stop SGLT2 inhibitor), hypotension/falls, euglycemic DKA symptoms

Cases 5, 6 and 7: Hyperthyroidism, Thyrotoxicosis and Thyroid Storm ★ Repeated case, also in 2022–2023, 2021, 2018–2019

Case details: Hyperthyroid patient: low TSH, high T4, high heart rate, weight loss. Refused surgery. Later presented to the ED with fever, tachycardia and high BP after stopping her thyroid medication.

Question	Answer
Q1. Diagnose from the given labs. What is the diagnosis?	• Hyperthyroidism/thyrotoxicosis: low TSH with high free T4 (and/or T3) • Causes: Graves disease (TRAb, ophthalmopathy, diffuse goiter), toxic nodular goiter, thyroiditis

Question	Answer
Q2. Signs and symptoms?	• Heat intolerance and sweating • Weight loss with a good appetite • Palpitations, tachycardia, tremor • Anxiety, insomnia, diarrhea • Exophthalmos and goiter (Graves)
Q3. The patient refused surgery. What pharmacological management will you start (agents and doses)?	• Methimazole 10–30 mg daily (PTU 50–150 mg q8h in the first trimester or MMI intolerance) • Propranolol 20–40 mg q6–8h for symptoms • Radioactive iodine is an alternative definitive option (not in pregnancy) • Duration of antithyroid drugs: 12–18 months
Q4. What will you monitor after starting the drug (safety and efficacy)?	• Efficacy: free T4/T3 and TSH every 4–6 weeks until euthyroid, then every 3 months; heart rate, weight • Safety: baseline CBC with differential and LFTs; agranulocytosis (sore throat, fever) and hepatotoxicity • Propylthiouracil: boxed warning for liver failure • Methimazole: teratogenic (aplasia cutis), cholestasis
Q5. Six months later she came to the ED with fever, tachycardia and high BP (stopped medication). Diagnosis? How do you manage it?	• Thyroid storm (Burch-Wartofsky score ≥45) • Propylthiouracil 500–1000 mg load then 250 mg q4h (or Methimazole 60–80 mg/day) • Iodine (Lugol 8 drops or Potassium iodide 5 drops q6h) at least 1 hour after the thionamide • Propranolol 60–80 mg q4–6h (or Esmolol) • Hydrocortisone 100 mg IV q8h • Cooling with Acetaminophen (no Aspirin); Cholestyramine 4 g q6h if needed • Treat the precipitant (stopped drugs, infection); ICU monitoring
Also be ready for (exam add-ons)	• Goals: normalize TSH and free T4 (hypothyroidism) or free T4/T3 (hyperthyroidism), relieve symptoms, prevent complications and relapse • Non-pharm: iodine intake awareness, smoking cessation (Graves ophthalmopathy; moderate to severe active orbitopathy: Teprotumumab, an anti-IGF-1R monoclonal antibody, or IV Methylprednisolone), sun/eye protection, avoid biotin before tests, adherence • First-line: hypothyroidism: Levothyroxine 1.6 mcg/kg/day (start 25–50 mcg in elderly or CAD), recheck TSH 6–8 weeks; hyperthyroidism: Methimazole 10–30 mg daily (12–18 months) + Propranolol 10–40 mg q6–8h; Radioiodine or surgery for definitive therapy • Pregnancy: hypothyroid: increase Levothyroxine 25–30% at once (two extra tablets/week), TSH <2.5, check every 4 weeks until week 20; hyperthyroid: Propylthiouracil in the first trimester, Methimazole after; avoid Radioiodine; postpartum return to pre-pregnancy dose • Renal/dialysis: Levothyroxine no adjustment (monitor TSH); Methimazole no adjustment • Liver: Propylthiouracil hepatotoxic; Methimazole cholestasis; baseline LFTs • HF/CAD: Levothyroxine start low (12.5–25 mcg); Propranolol caution in decompensated HF • GI: PPIs, calcium, iron, antacids, Sucralfate and cholestyramine reduce Levothyroxine absorption (separate by 4 h) • Counseling: timing, lifelong therapy, symptoms improve in 2–3 weeks and TSH in 6–8 weeks; Methimazole: agranulocytosis, rash, teratogenic (aplasia cutis); report palpitations or chest pain

Case 8: Mixtard Insulin Switching ★ Repeated case, also in 2022–2023, 2021, 2018–2019

Case details: Patient on Mixtard Insulin with uncontrolled A1C, planned for discharge on basal-bolus Insulin.

Question	Answer
Q1. How to convert the dose?	• Calculate TDD by adding all Mixtard units in 24 hours • Example: 30 units AM + 30 units PM = 60 units
Q2. How to divide the doses?	• Basal 50%: Glargine 30 units once daily • Bolus 50%: divided into 3 meals: Aspart 10 units before each meal • If hypoglycemia risk: reduce the starting TDD by 10–20%
Q3. How and when to take it?	• Basal: at the same time every day • Bolus: 0–15 minutes before meals; skip it if the meal is skipped • Adjust every 3 days by SMBG (fasting for basal, pre-meal and bedtime for bolus)
Q4. Patient education?	• Hypoglycemia: symptoms and the 15-15 rule; carry glucose tablets • Injection technique, site rotation, storage • SMBG before meals and bedtime; bring logs • Sick-day rules, sharps disposal, diet and exercise
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120

Question	Answer
	mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

2018–2019

Case 1: Type 1 DM Insulin Regimen ★ Repeated case, also in 2022–2023, 2021, 2018–2019, 2020

Case details: Type 1 DM, regimen and adjustment.

Question	Answer
Q1. Type 1 DM Insulin regimen, dosing and adjustment?	• TDD: 0.4–0.6 unit/kg/day • Basal 50%: Glargine or Detemir once daily; bolus 50% before meals • Insulin-to-carb ratio: 500 ÷ TDD; correction factor: 1800 ÷ TDD • Adjust: basal by 10–20% (or 2 units) every 3 days from fasting readings; adjust pre-meal doses from the next pre-meal reading • Monitor: SMBG 4+ times/day or CGM, A1C every 3 months, hypoglycemia
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware • Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months

Case 2: Uncontrolled DM with Frequent UTI

Case details: Uncontrolled DM with frequent UTI on Metformin only.

Question	Answer
Q1. What information do you ask from the patient?	• Adherence to Metformin and diet • Glucose logs, A1C, hypoglycemia, eGFR • UTI details: symptoms, culture results, previous treatment; urinalysis (glycosuria) • History of pancreatitis, thyroid cancer, weight and BMI, comorbidities (CKD, HF, ASCVD) • If adherent and uncontrolled: add a second agent; if not adherent: address barriers first • Choice here: GLP-1 RA (avoid SGLT2 inhibitors because of genitourinary infections)
Q2. Monitoring: when?	• A1C: every 3 months until at goal, then at least twice a year • Glucose: SMBG; urine ketones if ill • eGFR: at least yearly (every 3–6 months if eGFR 45–59; every 3 months if 30–44) • Vitamin B12 every 2–3 years on Metformin; folate if megaloblastic anemia • Annual: CBC, lipids, urine ACR, eye and foot exam
Also be ready for (exam add-ons)	• Goals: control glucose, prevent recurrent UTI and complications, preserve kidney function • Non-pharm: hydration, hygiene, glucose control, urinary tract evaluation (obstruction, retention), adherence assessment • First-line: add a GLP-1 RA (Semaglutide 0.25 mg weekly) or DPP-4 inhibitor (Linagliptin 5 mg); avoid SGLT2 inhibitors (genitourinary infection); Metformin continue

Question	Answer
	• Pregnancy: asymptomatic bacteriuria should be treated; Insulin, Cephalexin/Amoxicillin/Fosfomycin for UTI; avoid Nitrofurantoin near term, TMP-SMX first trimester, fluoroquinolones • Renal/dialysis: Nitrofurantoin avoid CrCl <30; Metformin eGFR rules • Counseling: UTI symptoms and early treatment, hydration, voiding habits, adherence, GLP-1 RA GI effects • Monitoring: A1C, eGFR, urinalysis/culture, B12 every 2–3 years

Case 3: DKA (Detailed) ★ Repeated case, also in 2025, 2022–2023, 2021

Case details: Detailed DKA case; potassium high at presentation.

Question	Answer
Q1. What is your assessment?	• DKA: hyperglycemia, ketosis, acidosis with a high anion gap, severity by pH and bicarbonate • Dehydration: deficit 6–10 L • Hyperkalemia from transcellular shift (acidosis, insulin deficiency) despite total body potassium depletion; serum K falls once Insulin is started
Q2. What are the risk factors in this patient to develop DKA?	• Insufficient or omitted Insulin • Infection • Acute illness: MI, stroke, pancreatitis • Drugs: steroids, SGLT2 inhibitors, antipsychotics, diuretics • New-onset type 1 diabetes
Q3. How will you start: with what, and the dose for both loading and infusion?	• First fluids: 0.9% NaCl 15–20 mL/kg in hour 1 • Then Insulin (if K ≥3.3): regular Insulin 0.1 unit/kg IV bolus, then 0.1 unit/kg/h (or 0.14 unit/kg/h without bolus) • Titrate hourly to lower glucose by 50–75 mg/dL/h
Q4. What about electrolytes: which one will you start and how much (K was high)?	• Potassium >5.2: none; recheck in 2 hours • Start 20–30 mEq/L once K 3.3–5.2 • Goal K 4–5 mEq/L • Phosphate only if <1 mg/dL
Q5. What will you monitor?	• Glucose hourly • BMP, bicarbonate, anion gap, venous pH every 2–4 hours • Potassium every 2 hours initially • Ketones (β-hydroxybutyrate), ECG, urine output, vitals, mental status
Q6. After 24 hours: glucose 10 mmol/L (180 mg/dL) and potassium 4.5. What will you do with Insulin and fluids?	• If DKA resolved (bicarbonate ≥15, pH >7.3, AG ≤12): start SC basal-bolus Insulin with a 1–2 hour overlap, then stop the drip • If not resolved: continue Insulin 0.02–0.05 unit/kg/h with dextrose 5% in 0.45% NaCl at 150–250 mL/h • Potassium: continue 20–30 mEq/L to keep 4–5 • Monitor: glucose every 1–2 hours, ketones, K
Also be ready for (exam add-ons)	• Goals: correct dehydration, hyperglycemia, ketoacidosis and electrolytes, treat the precipitant, then transition to SC Insulin and prevent recurrence • Non-pharm: ICU/step-down monitoring, NPO initially, strict intake/output, treat infection source, sick-day and ketone education, assess adherence barriers (cost, fear, technique) • First-line: 0.9% NaCl 15–20 mL/kg first hour; regular Insulin 0.1 unit/kg IV bolus then 0.1 unit/kg/h (only if K ≥3.3); K 20–30 mEq/L once K <5.2; Dextrose 5% when glucose <200; Bicarbonate only if pH <7.0; resolve: glucose <200 + 2 of HCO3 ≥15, pH >7.3, AG ≤12 • Pregnancy: DKA occurs at lower glucose levels (euglycemic) and is an emergency for the fetus: same protocol with fetal monitoring, left lateral position, K and phosphate care; Insulin is the only agent; SGLT2 inhibitors and GLP-1 RA stopped • Renal/dialysis: reduce Insulin needs (lower clearance), watch K and phosphate, careful fluids in oliguria; Metformin hold; Glibenclamide avoid • Liver: hypoglycemia risk (gluconeogenesis impaired); lactate interpretation • HF/CKD: fluids 250–500 mL/h guided by volume status; monitor for pulmonary edema • Counseling: never stop basal Insulin, sick-day rules, check ketones when glucose >250 or ill, SMBG 4×/day, hypoglycemia 15-15 rule, return precautions, SGLT2 inhibitors can cause euglycemic DKA • Monitoring: glucose hourly; BMP/venous gas every 2–4 h; K every 2 h at first; β-hydroxybutyrate; ECG; urine output

Case 4: Hypothyroidism (Direct) and Hyperthyroidism ★ Repeated case, also in 2022–2023, 2021, 2020

Case details: Direct hypothyroidism case, then a hyperthyroidism case.

Question	Answer
Q1. What is the assessment?	• Primary hypothyroidism: high TSH and low free T4 • Causes: Hashimoto thyroiditis (anti-TPO), post-thyroidectomy or radioiodine, drugs (Amiodarone, Lithium)

Question	Answer
Q2. What are the symptoms in this patient?	• Fatigue, cold intolerance • Weight gain, constipation • Dry skin, hair loss • Bradycardia, depression • Menstrual irregularity
Q3. Laboratory values?	• TSH: high (>4.5 mIU/L) • Free T4: low • Anti-TPO antibodies positive; lipids may be high
Q4. What medication, with dose (full regimen)?	• Levothyroxine 1.6 mcg/kg/day • Start 25–50 mcg/day in the elderly or with coronary disease; increase every 6–8 weeks • Monitor: TSH 6–8 weeks after each change
Q5. Counseling points?	• Take on an empty stomach 30–60 minutes before breakfast with water • Same time daily; do not switch brands without recheck • Separate from calcium, iron, antacids and PPIs by 4 hours • Report palpitations, chest pain or overtreatment symptoms
Q6. Patient asks: when will I start to see the effect?	• Symptoms: improve in 2–3 weeks • Full effect and TSH normalization: 6–8 weeks • Lifelong therapy in most patients
Q7. Hyperthyroidism: diagnosis, medications with doses, duration of treatment?	• Diagnosis: low TSH, high free T4/T3 (Graves with TRAb) • Methimazole 15–30 mg daily; Propranolol 10–40 mg q6–8h • Duration: 12–18 months, then reassess • Monitor: TFTs every 4–6 weeks, CBC and LFTs
Also be ready for (exam add-ons)	• Goals: normalize TSH and free T4 (hypothyroidism) or free T4/T3 (hyperthyroidism), relieve symptoms, prevent complications and relapse • Non-pharm: iodine intake awareness, smoking cessation (Graves ophthalmopathy; moderate to severe active orbitopathy: Teprotumumab, an anti-IGF-1R monoclonal antibody, or IV Methylprednisolone), sun/eye protection, avoid biotin before tests, adherence • First-line: hypothyroidism: Levothyroxine 1.6 mcg/kg/day (start 25–50 mcg in elderly or CAD), recheck TSH 6–8 weeks; hyperthyroidism: Methimazole 10–30 mg daily (12–18 months) + Propranolol 10–40 mg q6–8h; Radioiodine or surgery for definitive therapy • Pregnancy: hypothyroid: increase Levothyroxine 25–30% at once (two extra tablets/week), TSH <2.5, check every 4 weeks until week 20; hyperthyroid: Propylthiouracil in the first trimester, Methimazole after; avoid Radioiodine; postpartum return to pre-pregnancy dose • Renal/dialysis: Levothyroxine no adjustment (monitor TSH); Methimazole no adjustment • Liver: Propylthiouracil hepatotoxic; Methimazole cholestasis; baseline LFTs • HF/CAD: Levothyroxine start low (12.5–25 mcg); Propranolol caution in decompensated HF • GI: PPIs, calcium, iron, antacids, Sucralfate and cholestyramine reduce Levothyroxine absorption (separate by 4 h) • Counseling: timing, lifelong therapy, symptoms improve in 2–3 weeks and TSH in 6–8 weeks; Methimazole: agranulocytosis, rash, teratogenic (aplasia cutis); report palpitations or chest pain

Case 5: Insulin TDD and Basal-Bolus Conversion ★ Repeated case, also in 2022–2023, 2021, 2018–2019, 2020

Case details: Insulin TDD calculation and conversion between insulins.

Question	Answer
Q1. Calculate TDD, basal-bolus regimen, and conversion between insulins.	• TDD: 0.4–0.6 unit/kg/day (T1DM); sum current Insulin units for conversion • Basal-bolus: 50% basal (Glargine), 50% bolus divided before meals • NPH to Glargine: same total units, once daily (reduce by 20% if hypoglycemia risk) • Premixed (Mixtard) to basal-bolus: same TDD split 50:50 • Monitor: SMBG and adjust every 3 days
Also be ready for (exam add-ons)	• Goals: A1C <7% (individualize), no severe hypoglycemia, stable fasting 80–130 mg/dL and postprandial <180 • Non-pharm: carbohydrate counting, consistent meal timing, 150 min/week activity (adjust Insulin), weight management, SMBG/CGM, education, foot/eye/kidney screening, sick-day rules • First-line: T1DM TDD 0.4–0.6 unit/kg/day (half basal Glargine/Detemir, half bolus Aspart/Lispro before meals); T2DM basal 10 units or 0.1–0.2 unit/kg then titrate by 2 units every 3 days; conversions: NPH or Mixtard TDD split 50:50 basal:bolus (reduce 10–20% if hypoglycemia risk) • Pregnancy: Insulin is the drug of choice (Aspart, Lispro, Detemir, NPH, Regular); stop SGLT2 inhibitors, GLP-1 RA, DPP-4 inhibitors, ACE inhibitors/ARBs and statins; targets fasting 70–95, 1 h <140, 2 h <120 mg/dL; Insulin needs fall in the first trimester and rise from 16–20 weeks; fall again after delivery • Renal/dialysis: lower Insulin requirement (reduce 25–50% in HD); A1C unreliable (use CGM/SMBG); hypoglycemia risk • Liver: reduced gluconeogenesis and Insulin clearance: lower doses • HF/elderly: avoid Pioglitazone; A1C goal <8% if frail or hypoglycemia unaware

Question	Answer
	• Counseling: technique (new needle, rotate sites, 2 inches from the umbilicus, no lumps), storage (unopened in fridge; in-use pen 28–30 days at room temperature), timing (bolus 0–15 min before meals, skip if no meal), hypoglycemia (symptoms, 15 g carbohydrate, recheck in 15 min, glucagon), driving, alcohol, sharps disposal, weight gain • Side effects: hypoglycemia, weight gain, lipohypertrophy, injection-site reaction; monitoring SMBG before meals and bedtime, A1C every 3 months